

Time Series Analysis of Stock Indices and Cryptocurrencies Using Econometric and Machine Learning Models Before and During the COVID-19 Pandemic

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Abstract—

Forecasting financial market prices is a challenging task since prices exhibit a high level of fluctuations, especially during the crisis period, for instance, like that of COVID-19. In this study, various models like ARIMA, GARCH, and LSTM models were tested on both global stock indices as well as cryptocurrencies such as S&P 500, ASX 200, Ibovespa, Bitcoin, Ethereum, and Dogecoin. The main novelty of this study is not only to evaluate different models but also to assess the change in their performances due to shifts in market conditions between the pre-COVID and COVID periods. The daily closing prices' dataset was obtained from Yahoo Finance and pre-processed through time-based preprocessing techniques such as data cleaning, scaling, log-return, sliding window, and chronological train-test splits. Performance evaluations of models were based on metrics such as RMSE, MSE, and R-squared. According to findings, the ARIMA model outperformed the LSTM model on the vast majority of financial assets, whereas the GARCH model helped to identify the behaviour of volatility mostly within cryptocurrency assets.

Keywords—Time series Forecasting, ARIMA, GARCH, LSTM, Financial Markets, Volatility

I. INTRODUCTION

Financial markets generally produce complex time series data which is influenced by volatility, nonlinearity and sudden structural changes. Stock indices, such as the S&P 500, ASX 200, and Ibovespa depict overall market condition, whereas cryptocurrencies such as Bitcoin, Ethereum and Dogecoin are the latest innovations with highly fluctuating values. The COVID-19 pandemic made it harder to predict what would

happen in the market. There was a lot of uncertainty during those periods (Baker et al., 2020) .

Traditional forecasting models such as ARIMA and GARCH are still widely used in financial time series because they can capture trend, time dependence and volatility behaviour. However, recent studies show that deep learning models, LSTM-based models, are increasingly used because they can learn nonlinear patterns in stock and cryptocurrency data (Yao & Yan, 2024) (Huang et al., 2024). Simultaneously, other research articles that compare deep learning with traditional ones have found that deep learning does not always automatically outperform traditional models, and performance depends on the asset type, volatility level, data structure and evaluation method (Taslim & Murwantara, 2024); (Mehtarizadeh et al., 2025).

Although many studies compare forecasting models, a clear research gap still remains. Almost all the existing research focuses on one asset class, one market period or one modelling approach. Such as: Some studies focus primarily on stock market prediction, mainly on crypto volatility or specific deep learning models (Gupta, Samraj; Nachappab, Sanchal; Paramanandhama, Nirmala, 2025) (Huang et al., 2024). Therefore, there is limited evidence on how ARIMA, GARCH and LSTM perform across both traditional stock indices and highly volatile cryptocurrencies under different market conditions.

Hence, this paper compares ARIMA, GARCH and LSTM models across selected stock indices and cryptocurrencies in the pre-COVID and COVID periods. The motivation for selecting these three models is that they explain a different part of financial behaviour: ARIMA captures linear price movements, GARCH captures volatility clustering, and LSTM captures nonlinear sequential patterns. The main contribution of this research is its cross-asset and regime-based comparison, which portrays how model performance alters between normal and crisis market conditions. This can clearly show whether classical statistical models remain

consistent compared with deep learning approaches in financial forecasting.

II. RELATED WORK

Statistical models based on the ARIMA model have been extensively applied in financial time series forecasting as they can capture linear relationships, autocorrelation and short term time-dependent patterns after differencing. The classical literature on time series supports the application of ARIMA to structured sequences of data, when these sequences can be converted to a more stationary form (Box et al., 2015). Nevertheless, ARIMA may be constrained in situations where financial data exhibits strong nonlinearity, sudden structural shifts or unsteady variance which are common in crisis periods.

In order to deal with changing variance in financial markets, GARCH models have been widely used to deal with volatility modelling. GARCH was created to model conditional heteroskedasticity when volatility varies over time and depends on past shocks and past variance (Bollerslev, 1986). This is useful in financial forecasting since stock indices and other cryptocurrencies tend to perform volatility clustering, particularly during uncertain times, such as COVID-19. Recent research undertaking Bitcoin volatility studies, has compared traditional econometric models to machine learning models and have found that model performance can vary depending on the forecast horizon and market condition (Huang et al., 2024).

Due to their ability to discover nonlinear trends and long-term relationships in sequential data, deep learning models, in particular LSTM, have become popular in financial forecasting in recent years. Some studies have demonstrated that LSTM-based models could potentially enhance their forecasting performance in stock market and cryptocurrency prediction when the dataset is large enough and the model has been properly tuned (Yao & Yan, 2024) (Gupta, Samraj; Nachappab, Sanchal; Paramanandhama, Nirmala; 2025). However, the comparative studies also indicate that LSTM is not always better than traditional models as its performance can be dependent on the quality of data, the degree of volatility, the way features are prepared and hyperparameters selected (Taslim & Murwantara, 2024).

Overall, the current studies have compared ARIMA, GARCH and LSTM in various financial forecasting settings, such as predicting the stock market and predicting the volatility of cryptocurrencies. However, much of the available literature is focused on one or another type of asset, one market situation or one model family. The study that is still missing is a more clearer cross-asset and regime-based comparison which evaluates both traditional stock indices and highly volatile cryptocurrencies both before and during the COVID-19 epidemic. In this study, the gap is bridged by comparing ARIMA, GARCH and LSTM under normal and crisis market conditions, which helps understand how the stability of models varies in normal market conditions and during crises.

III. DATASET AND PREPROCESSING

a) Data Source and Asset Classification

This financial data was integrated directly from Yahoo Finance via an API, utilizing the Python yfinance library, which withdraws data from the Yahoo Finance website (Gunawan et al., 2024).

The datasets comprise multi-asset and cross-market financial data, representing the daily closing prices for each asset. The assets are classified into two distinct categories low volatility assets (S&P 500) from the USA (Ticker: "^GSPC"), Ibovespa, from Brazil (Ticker: "^BVSP"), and ASX 200 from Australia (Ticker: "^AXJO"), and high volatility assets, represented by Bitcoin (Ticker: "BTC-USD"), Ethereum (Ticker: "ETH-USD"), and Dogecoin (Ticker: "DOGE-USD").

The stock indices under consideration in this paper cover a long historical period from 2000 to 2024. Meanwhile, cryptocurrencies are considered young financial assets which were developed due to advances in computer technologies. After the global financial crisis of 2008-2009, the cryptocurrency industry appeared in 2009 with the invention of Bitcoin and blockchain technology.

b) Data cleaning.

The cryptocurrency datasets did not show problems with missing values. As the missingness of the Indices datasets (Figure 1) was well distributed over the dataset and represented less than 5 % of the total data, it was opted to use the dropna function to remove any null values. Thus, the dataset was rendered structurally consistent for the LSTM and ARIMA models while preserving the chronological integrity of the financial records.

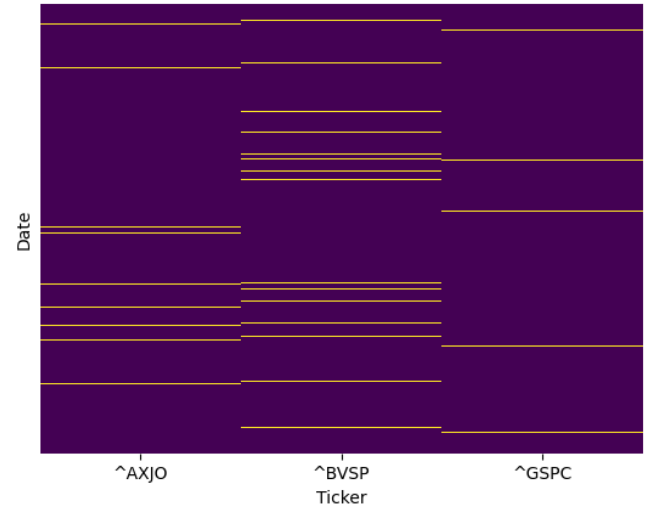


Figure 1. Missingness for Indices dataset.

c) Regime Split

The indices span from 2000 to 2024 and the cryptocurrency, as it is a more recent financial asset that was only made possible by the evolution blockchain technology, officially began in 2009 with the launch of Bitcoin and blockchain technology. It gained momentum after the 2008/2009 subprime mortgage crisis (Malik, 2016). Data-wise, the earliest data available via Yahoo Finance begins on September 17, 2014.

As other cryptocurrencies were developed in subsequent years, only Bitcoin has a tracking history dating back to this 2014 period. For this study, Ethereum (ETH) and Dogecoin (DOGE) data begins on November 9, 2017.

Covid-19 was an infectious disease caused by SARS-CoV-2 and it started in Wuhan, China, in late 2019. The first

case registered in the EUA was in late January 2020. The World Health Organization declared a global health emergency in March 2020 and the pandemia officially finished more than three years later, in May 2023 (Northwestern Medicine, 2025).

This study will consider that the start and end dates for pandemic period as 1st January 2020 and 31st December 2023, as there was still volatility and uncertainty before and after the official dates.

In both indices and cryptocurrency, the data was partitioned into four distinct datasets based on their respective start dates and two primary timeframes: Pre-Covid (from 01/01/2000 to 31/12/2019) and During Covid (from 01/01/2020 to 31/12/2023). The distribution of this data is detailed in Table 1.

Indice	Pre-covid	Covid
	S&P 500 (USA)	S&P 500 (USA)
	Ibov (Brazil) and ASX 200 (Australia)	Ibov (Brazil) and ASX 200 (Australia)
Cryptocurrency	BTC	BTC
	ETH	ETH
	DOGE	DOGE

Table 1: Distribution of all datasets

Figures 2 and 3 below show the volatility plots (time series plots) for the Indices and Cryptocurrencies, respectively. These plots were made possible through feature scaling (normalization). As the assets operate at different orders of magnitude, the values were scaled to a range between 0 and 1 to allow for a direct comparison.



Figure 2. Volatility plots for the indices..

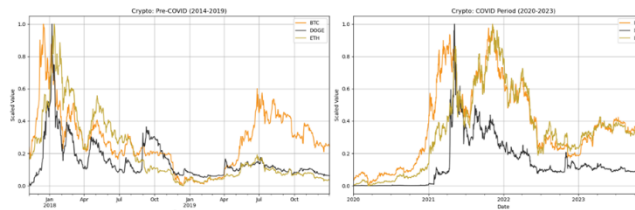


Figure 3. Volatility plots for the cryptocurrency.

d) Train and Test splitting for reproducibility

In order to maintain reproducibility of our results, a matter that is very important for later peer review, it was implemented fixed random seeds. This approach ensured that the chronological data split would remain consistent and that the stochastic elements of the LSTM model, such as weight initialization and the dropout layers, which randomly deactivate neurons to prevent overfitting, would produce identical results across multiple executions.

We understand that a proper approach would be to run the code multiple times and then get the average of the metrics.

This option will not be approached due to limited computing power, as each run is taking almost 60 minutes to be completed for an average computer.

For all the models, a chronological train/test split was implemented using an 80% training and 20% testing ratio. This specific splitting method is very important in time-series analysis as the temporal order of observations is maintained.

e) Pre-processing steps according to each model

The last preprocessing step took place to satisfy the assumptions of both statistical and deep learning architectures. For the ARIMA and GARCH models, the closing prices were converted into log-returns to stabilize the mean and variance. For the LSTM, features were normalized using MinMaxScaler to a range from 0 to 1. This scaling is essential to prevent large absolute price values from destabilizing the gradient descent process, thereby facilitating efficient neural network convergence.

The LSTM data was also restructured into a supervised learning format using a 30-day sliding window (lookback). This process allows the deep learning model to capture vital temporal dependencies and non-linear patterns by using a sequence of past observations to predict the subsequent price point. Table 2 outlines the details that each model needs as preprocessing steps.

Preprocessing Step	ARIMA	GARCH	LSTM
Log>Returns	Yes (Primary)	Yes (Primary)	No*
MinMaxScaler	No	No	Yes (Required)
Sliding Window	No	No	Yes (Required)

Table 2: Pre-processing steps for each model

f) Model Parameters

The statistical ARIMA (Autoregressive Integrated moving Average) model is defined as $ARIMA(p, d, q)$ and it has three main parameters: p , the number of autoregressive terms; d the degree of differencing involved; and q the order of moving average (Noble, n.d.).

It was implemented $ARIMA(5,1,0)$ model using a walk-forward validation approach. In this configuration, it was selected 5 autoregressive terms ($p=5$) to capture short-term momentum and a single order of differencing ($d=1$) to ensure the stationarity of financial price levels.

The GARCH model is defined as $GARCH(p, q)$, where p is the number of previous variance and q is the number of previous residual. (Liu J. , 2018). For this study, $GARCH(1,1)$ model was integrated to estimate conditional volatility, employing one lag of residual variance and one lag of squared residuals.

The deep learning model LSTM (Long Short-Term Memory) was implemented to capture non-linear dependencies within a 30-day lookback window. The architecture consisted of two hidden LSTM layers with 64 and 32 units respectively, separated by Dropout layers (0.2) to mitigate overfitting by randomly deactivating 20% of the neurons during training. The model was optimized using the Adam algorithm to minimize Mean Squared Error (MSE) using a maximum of 20 epochs. An early stopping mechanism with a patience of 3 was enforced to halt training once the validation loss function stabilized.

IV. METHODOLOGY

The methodology of this work comprehends a comparative framework to address the research gap in forecasting performance between traditional statistical models and deep learning architectures across distinct financial regimes.

The evaluation and final results of models are quantified using three core metrics: RMSE (Root Mean Squared Error), R-squared Coefficient, and MSE (Mean Squared Error). Through this multi-regime comparison, the research analyses whether deep learning models maintain success across various volatility environments or require specific tuning to individual market regimes.

The models were run following Table 1 shown previously. Therefore, each one of the 12 datasets will have a particular chart showing the actual price (black lines), ARIMA forecast (dashed blue line) LSTM forecast (orange line) and volatility band (Gray area).

V. EXPERIMENTAL SETUP

- Programming: Python
- Libraries: Pandas, Statsmodels, TensorFlow, Arch
- Metrics: RMSE, MSE, R^2
- Validation: Chronological train-test split

VI. RESULTS AND DISCUSSION

A. Volatility plots and predicting curves

i. Indices During Pre-Covid

Figures 4 present the volatility plots and corresponding prediction curves for the pre-COVID period, using a 20% training split. The analysed indices include the S&P 500, from the USA (Ticker: " $^{\wedge}$ GSPC") , Ibovespa, from Brazil (Ticker: " $^{\wedge}$ BVSP"), and ASX 200 from Australia (Ticker: " $^{\wedge}$ AXJO").

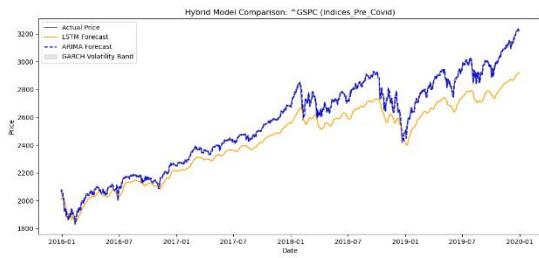


Figure 4. Volatility plots for the 20% training split along with their respective prediction curves for the S&P 500, from the USA (Ticker: " $^{\wedge}$ GSPC") during pre-COVID times.

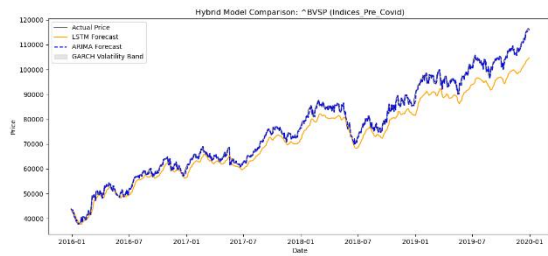


Figure 5. Volatility plots for the 20% training split along with their respective prediction curves for the Ibovespa, from Brazil (Ticker: " $^{\wedge}$ BVSP") during pre-COVID times.

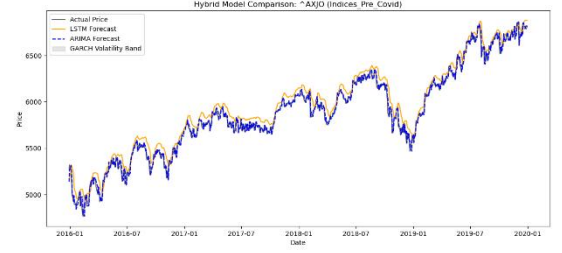


Figure 6. Volatility plots for the 20% training split along with their respective prediction curves for the ASX 200 from Australia (Ticker: " $^{\wedge}$ AXJO") during pre-COVID times.

ii. Indices During Covid

Figures 7 through 9 present the volatility plots and corresponding prediction curves for the COVID period, using a 20% training split. The analysed indices include the S&P 500, from the USA (Ticker: " $^{\wedge}$ GSPC") , Ibovespa, from Brazil (Ticker: " $^{\wedge}$ BVSP"), and ASX 200 from Australia (Ticker: " $^{\wedge}$ AXJO").

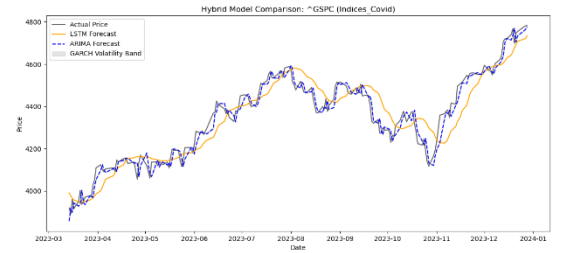


Figure 7. Volatility plots for the 20% training split along with their respective prediction curves for the S&P 500, from the USA (Ticker: " $^{\wedge}$ GSPC").

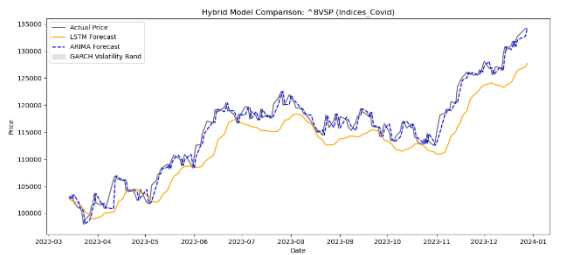


Figure 8. Volatility plots for the 20% training split along with their respective prediction curves for the Ibovespa, from Brazil (Ticker: " $^{\wedge}$ BVSP") during COVID times.

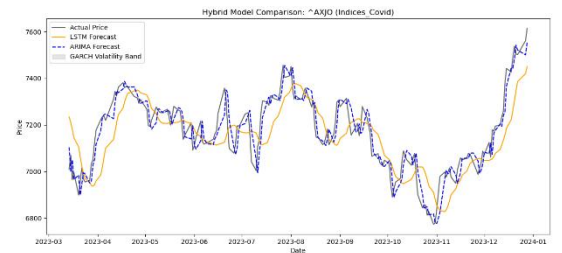


Figure 9. Volatility plots for the 20% training split along with their respective prediction curves for the ASX 200 from Australia (Ticker: "^AXJO") during COVID times.

iii. Cryptocurrency during preCovid

Figures 10 through 11 present the volatility plots and corresponding prediction curves for the pre-COVID period, using a 20% training split. The analyzed cryptocurrencies include Bitcoin (Ticker: "BTC-USD"), Ethereum (Ticker: "ETH-USD"), and Dogecoin (Ticker: "DOGE-USD").

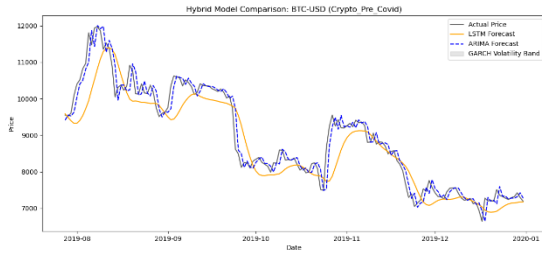


Figure 10. Volatility plots for the 20% training split along with their respective prediction curves for Bitcoin (Ticker: "BTC-USD") during pre-COVID times.

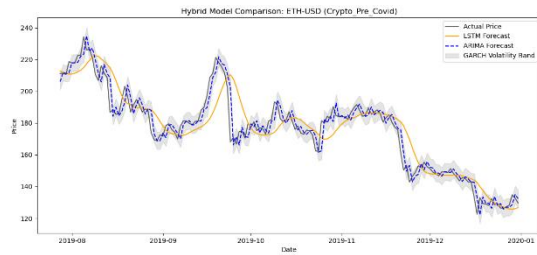


Figure 11. Volatility plots for the 20% training split along with their respective prediction curves for Ethereum (Ticker: "ETH-USD") during pre-COVID times.

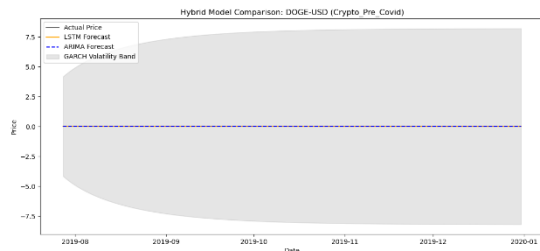


Figure 12. Volatility plots for the 20% training split along with their respective prediction curves for Dogecoin (Ticker: "DOGE-USD") during pre-COVID times.

iv. Cryptocurrency during COVID

Figures 13 through 15 present the volatility plots and corresponding prediction curves for the COVID period, using a 20% training split. The analyzed cryptocurrencies include Bitcoin (Ticker: "BTC-USD"), Ethereum (Ticker: "ETH-USD"), and Dogecoin (Ticker: "DOGE-USD").

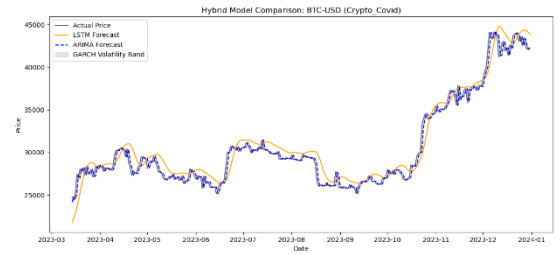


Figure 13. Volatility plots for the 20% training split along with their respective prediction curves for Bitcoin (Ticker: "BTC-USD") during COVID times.

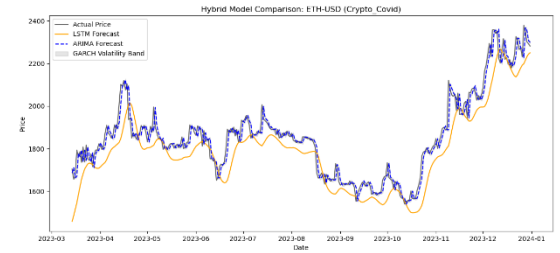


Figure 14. Volatility plots for the 20% training split along with their respective prediction curves for Ethereum (Ticker: "ETH-USD") during COVID times.

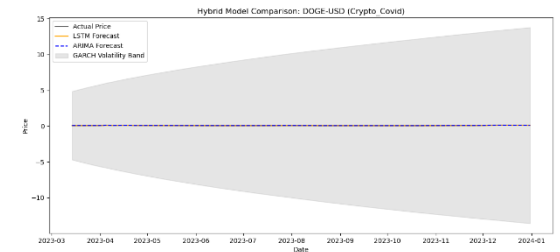


Figure 15. Volatility plots for the 20% training split along with their respective prediction curves for Dogecoin (Ticker: "DOGE-USD") during COVID times.

B. Model Performance

Table 2 and Table 3 show the the core metrics, RMSE (Root Mean Squared Error), R-square Coefficient, and MSE (Mean Squared Error), for ARIMA and LSTM models per asset type.

Asset	Regime	RMSE_ARIMA	RMSE_LSTM	R2_ARIMA	R2_LSTM	MSE_ARIMA	MSE_LSTM
^GSPC	Indices Pre_Covid	20.99	134.17	0.996	0.837	440.60	18,002.84
^GSPC	Indices Covid	35.28	73.85	0.969	0.865	1,244.87	5,453.40
^BVSP	Indices Pre_Covid	1,012.42	4,738.94	0.997	0.934	1.02E+06	2.25E+07
^BVSP	Indices Covid	1,228.77	3,817.51	0.976	0.773	1.51E+06	1.46E+07
^AXJO	Indices Pre_Covid	43.68	96.96	0.992	0.958	1,908.29	9,401.80
^AXJO	Indices Covid	53.93	106.62	0.887	0.559	2,908.11	11,367.90

Table 2. RMSE, R-squared and MSE for ARIMA and LSTM models during pre-Covid and COVID periods for stock Indices.

Asset	Regime	RMSE_ARIMA	RMSE_LSTM	R2_ARIMA	R2_LSTM	MSE_ARIMA	MSE_LSTM
BTC-USD	Crypto Pre_Covid	274.79	543.38	0.96	0.84	7.55E+04	2.95E+05
BTC-USD	Crypto Covid	658.44	1,269.77	0.98	0.94	4.34E+05	1.61E+06
DOGE-USD	Crypto Pre_Covid	7.50E-05	1.12E-04	0.91	0.80	5.62E-09	1.26E-08
DOGE-USD	Crypto Covid	2.57E-03	5.15E-03	0.94	0.75	6.58E-06	2.65E-05
ETH-USD	Crypto Pre_Covid	6.19	11.29	0.94	0.80	38.30	127.43
ETH-USD	Crypto Covid	43.90	96.84	0.95	0.74	1,927.56	9,377.52

Table 3. RMSE, R-squared and MSE for ARIMA and LSTM models during pre-Covid and COVID periods for cryptocurrency.

VII. ANALYSIS

The results in Table 2 and Table 3 suggest that ARIMA produced more consistent forecasting performance than LSTM across most of the selected assets and time periods in this study. ARIMA generally recorded lower RMSE and MSE values and higher R^2 scores, which indicates that it was able to capture the main price movements more effectively under the selected dataset and model settings. This finding is consistent with the continued use of ARIMA for modelling linear and time-dependent patterns in financial time series (Taslim & Murwantara, 2024). However, this result should not be interpreted as evidence that ARIMA is always superior to deep learning models, as model performance can vary depending on the dataset, asset type, volatility condition and parameter tuning.

For the stock indices, ARIMA showed relatively stable performance across both pre-COVID and COVID periods. The strongest index result was observed for the Brazilian Ibovespa during the pre-COVID period, where ARIMA achieved a high R^2 value. This suggests that the model was able to explain a large proportion of variation in that index during a comparatively more stable market period. In contrast, LSTM showed weaker performance for some index datasets, especially during the COVID period for the ASX 200. This may be because sudden market shocks and structural breaks during crisis periods make it harder for neural network models to learn stable patterns without deeper tuning or additional features (Liu P. , 2022).

For the cryptocurrencies, ARIMA showed good results during the COVID periods. This result indicates that despite of how volatile the cryptocurrencies market is, ARIMA can still capture some short term trend patterns. Cryptocurrencies markets are very speculative and often shows nonlinear characteristics, volatility clustering and other sensitivity to external factors (Corbet et al., 2019) (Huang et al., 2024). So, the results should only be taken in consideration as ours experiment and not to all the cryptocurrency forecasting.

The LSTM model showed mixed performance across the tested datasets. Although LSTM models are designed to capture nonlinear relationships and long-term dependencies, their effectiveness often depends on data size, preprocessing, hyperparameter selection and model architecture (Gupta, Samraj; Nachappab, Sanchal; Paramanandhama, Nirmala;, 2025). In this study, LSTM appeared less stable for some assets, such as Dogecoin in the pre-COVID period. This may be due to the noisy and speculative nature of the asset, which can make sequential patterns harder to learn. There are many research paper that has found LSTM as the better performing models in some financial forecasting task but it doesn't always outperform traditional statistical models (Taslim & Murwantara, 2024) (Mehtarizadeh et al., 2025).

Overall from our research , traditional statistical models are still very useful in analysing financial time series when the data's are strong time dependent patterns. From our research we found that the performance of our models depends on the asset class, market regime, preprocessing methods and model architecture. These evidence provides support to our decision to compare the three models ARIMA, GARCH and LSTM across the stock indices and cryptocurrencies before and during the COVID 19 and rather than picking one model and one factor regime.

VIII. CONCLUSION

This research compared ARIMA, GARCH and LSTM models forecasting in pre-COVID and COVID market periods for selected indices and cryptocurrencies. The purpose of this study was to evaluate models as well as understand how models performed under different market conditions and various asset classes.

From the results, ARIMA gave lower RMSE and MSE values and stronger R^2 scores, showing consistent performance across multiple tested datasets. This justifies the further applicability of old statistical models to financial time series forecasting, especially when the data has clearly defined time-dependent patterns (Box et al., 2015) (Taslim & Murwantara, 2024). Nevertheless, these findings must be viewed in the context of the chosen assets of this study, its time span, preprocessing methods and models.

The LSTM model gave some mixed results. LSTM performance showed it is more sensitive to volatile, data structure and model tuning, even though it is designed to capture nonlinear and long term sequential patterns. This suggests that deep learning models might need more hyperparameter optimization, additional market characteristics or hybrid designs to perform consistently across different financial regimes (Liu P. , 2022) (Mehtarizadeh et al., 2025).

The comparison between COVID and pre-COVID period provided an insightful understanding that model reliability may shift when markets shift between relatively stable markets to high-uncertainty periods. GARCH also showed importance in this context as it described volatility behaviour particularly in highly volatile cryptocurrency markets (Bollerslev, 1986) (Huang et al., 2024). Overall, the results indicate that it is not possible to conclude a single model as the best. Rather, the performance following forecasting is dependent on the type of asset, the degree of volatility, the market regime and modelling strategy. This study contributes by providing a cross-asset and regime-based comparison of ARIMA, GARCH and LSTM, showing the practical value of testing both traditional and deep learning models under normal and crisis market conditions.

IX. FUTURE WORK

For this research, the models we used historical price data and simple settings. For better prediction, we can use more advanced methods and more datasets. We can use hybrid models like ARIMA & GRACH or ARIMA & LSTM together. Using such models will help identify both patterns and complex behaviour at the same time. These models might make predictions more accurate, especially when markets are unstable.

Since financial markets are driven by many other factors like past prices, world affairs, investor sentiment, technology, and government policies, etc. So, we can add macroeconomic variables or an analysis of the sentiment from news sources and social media, which will make the models more realistic and powerful. These models can react much better to the unexpected changes like COVID-19 and other pandemics.

The LSTM model in our research showed weak performance. For this, we can focus more on hyperparameter optimization, training for longer periods, larger dataset to fully utilize the capability of deep learning models. We will also try

other techniques like grid search, dropout layers and other different model architectures to improve the performance.

Lastly, future research can test models on time periods, like short-term and long-term predictions and use more financial assets. This will help us see how well each model works in situations and make results more reliable.

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